

Answer Key — Light: Mirrors and Lenses

Final answers with brief reasons.

Objective

- Q1 **(b) convex** — an outward-curving mirror is convex.
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- Q2 **(a) erect and diminished** — convex mirrors give erect, diminished images.
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- Q3 **(a) $i = r$** — angle of incidence = angle of reflection.
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- Q4 **(b) converge** — a concave mirror converges parallel rays.
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- Q5 **(b) convex lens** — a magnifying glass is a convex lens.
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- Q6 **(a) True** — lenses transmit light; mirrors reflect it.
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- Q7 **middle** — convex lenses are thick in the middle.
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- Q8 **convex** — side mirrors are convex for a wide erect view.
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Short answer

- Q9 (1) $i = r$ (angle of incidence = angle of reflection). (2) The incident ray, normal and reflected ray lie in the same plane.
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- Q10 Concave: torch reflector / dentist's mirror (concentrates light / enlarges). Convex: side-view/road-safety mirror (wide erect view).
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- Q11 Convex: thicker in the middle, converges light; concave: thicker at edges, diverges light.
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Long / application

- Q12 A convex mirror always gives an erect, diminished image and a wide field of view, so drivers see more traffic behind — ideal for side mirrors. A concave mirror converges light, so placing the bulb suitably produces a strong parallel beam — ideal for a torch/headlight reflector.
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- Q13 A concave mirror converges the Sun's parallel rays to a point, concentrating heat enough to burn paper. A convex lens also converges light; held close over print it forms an enlarged, erect image, so the letters look bigger. Both converge light — one by reflection, the other by transmission.
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